



MINING DEALER BEST PRACTICE

Barrel Port Plate Revision Tool

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

Currently at Borusan Makina ve Güç Sistemleri we are obliged to outsource for existing barrel port plate revision operations. However, when the components go to the external supplier, both the offer period and the deadline are experiencing major disruptions. This has been and continues to be an ongoing issue with the external supplier, and we believe it could apply to other parts of the world as well. This outsourcing also increases the cost of the value-added business reflected to the customer. And, due to the lack of tools, as external procurement is not under our control, we cannot follow the quality processes, meaning appropriate equipment and workforce adjustment cannot be performed correctly.

Due to this situation, we will utilize common shop materials and establish a tool to perform these operations ourselves, rather than outsourcing. This will increase efficiency and profitability, as well as adherence to quality processes.

2.0 Best Practice Description

This tool consists of assembled parts from different shop equipment attached to a drill. It can be used to revise the working surfaces of barrel and port plates and make them compatible with each other.

There was a drill sitting idle in the shop and a technician pulled it out to make it usable and turn it into a lathe. The parts shown below in the implementation steps are mounted onto the drill. The lathe chuck was made to 250 mm dimensions with 3 legs. The spring on the tool is approximately equal to the pressure applied by hand. The barrel and port plate are adjusted to the circular center with the interchangeable center. Any necessary modifications can be performed to revise the barrel port plate. The drill converted to a lathe by the mounted parts is shown below.

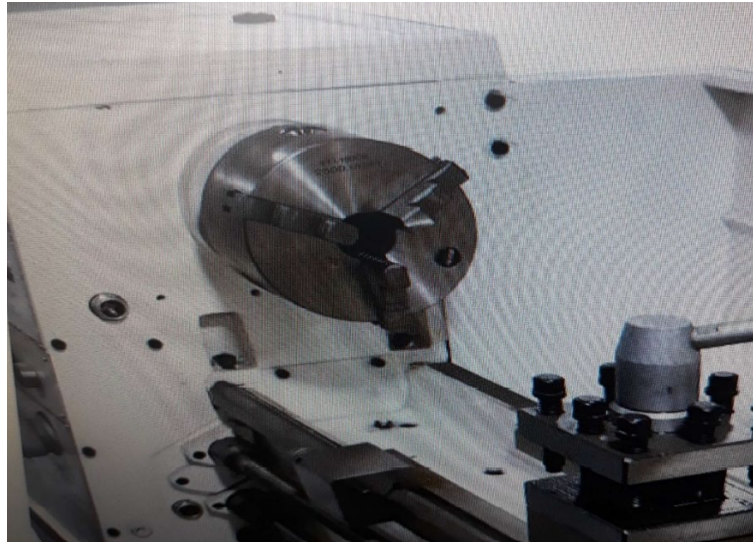


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3.0 Implementation Steps

1. Setup the drill.
2. Make the lathe chuck to the dimensions specified.



Lathe chuck with specified dimensions.

3. Attach the lathe chuck to the drill press.
4. Place the barrel on the drill by assembling it with the spring, which can be closed with hand pressure, and changing the lower center if needed.



5. Mount the port plate to the drill bit with the apparatus.

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Interchangeable centers for the lathe.



Barrel port plate with an interchangeable center and mounting apparatus.

6. Use paste for proper labbing at low speeds.

4.0 Benefits

- Standardizes and adheres to quality processes;
- Increases profitability by reducing outsourcing;
- Reduces cost to the customer;
- Improves efficiency by decreasing lost time during the supply waiting period and increasing quality;
- Prevents disruptions in the offer period and deadline.

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5.0 Resources Required

Item	Cost
Material	\$200
Labor	\$100
Total	\$300

6.0 Supporting Attachments / References

N/A

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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